

Real-time drone detection and recognition by acoustic fingerprint

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Abstract— In recent years, one of the important and interesting tasks has become the protection of civilian and military objects from unmanned aerial vehicles (UAVs) carrying a potential threat. To solve this problem, it is required to detect UAVs and activate protective systems. UAVs can be represented as aerodynamic objects of the monoplane or multicopter type with acoustic fingerprints. In this paper we consider algorithm for UAV acoustic detection and recognition system. Preliminary results of analysis of experimental data show effectiveness of proposed approach.

Keywords— Drone, UAV, acoustic signature, correlation

I. INTRODUCTION

UAVs represent a rapidly growing area of technology with applications in private and military sector. In recent years UAVs popularity have been growing rapidly due to decrease of size and weight, simplification of control system and wide availability. However, there have also been numerous cases of illegal use of drones with severe consequences (see, for example, [1]). Therefore, flight authorities working on regulation of drone operation. However, regulation cannot prevent criminal or terroristic actions, as well as accidents. Thus, it is necessary to simultaneously carry out the tasks of detecting, tracking and interdicting of drones. This task seems to be quite complicated and can be solved in various ways, including visual observation and detection of radio frequency signals with machine learning techniques. Recently several review papers have been published providing the advantages and disadvantages of drone detection methods [2-5]

Another promising drone observation technique is acoustic signature analysis [6-8]. This approach is based on the similarity of the spectral components of many commercially available UAVs. In the ideal case it is necessary to determine not only the presence of the drone, but its manufacturer and model. In this case, external noise (when analyzing in urban conditions), as well as the complex spectrum of oscillations created by drones, have a significant impact on the operation of analysis methods. An important criterion is also the response time of the detection algorithm. In modern conditions (taking into account the speed of movement of drones), it is necessary to achieve high accuracy of operation in several seconds. In this paper, we present preliminary results on testing an algorithm that satisfies these conditions. It should also be added that to improve the detection accuracy, the analysis is carried out in the frequency and time domains.

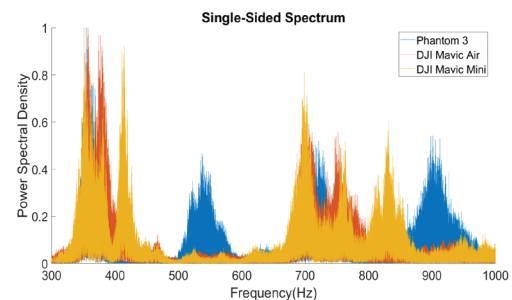


Fig. 1. Drones sound power density spectra: DJI Phantom 3 (blue), DJI Mavic Air (red), DJI Mavic Mini (yellow).

II. EXPERIMENTAL DATA ANALYSIS

A. Initial data and problem statement

To identify the characteristic features of information acoustic signals and creation of their adequate mathematical model, which is necessary at the stage of developing detection and recognition algorithms, we calculated power spectral density (PSD) of the UAV's acoustic signature. A typical implementation of the PSD in this case contains narrow-band spectral components of the fundamental tone, its harmonics and a wide-band noise component. To obtain the spectra, studio recordings of drone engines at rest with a length of 120 to 180 seconds with a sampling rate of 44100 Hz were used. (Fig. 1). The number of harmonics of the fundamental tone of the UAV is from 10 to 40, but with an increase in the distance to the UAV, due to absorption sound in the atmosphere, high-frequency harmonics are significantly attenuated to the level of background noise and therefore their inappropriate to include in the preprocessing. The pitch frequency of automobile engines is significantly lower than the rotational speed of aircraft engines and is in the range from 40 to 80 Hz, therefore, they can be distinguished from the sound trail of UAVs by the lower pitch frequency. This difference is characteristic distinctive feature.

B. Algorithm description

The most important task in UAV detection is the choice of the data processing step to ensure a high probability of correct recognition against the background natural and artificial noises. In particular, the algorithm includes several stages.

Input audio signal is continuously recorded and each chunk (fragment) of the signal is processed to conclude if one of the searched objects exist there.

The first stage of searching algorithm is filtering the searched objects by the amplitude spectrum of the recorded signal. We know that each of the searched objects has spectral components in determined diapasones. Each diapason is described by three parameters: begin frequency (f_b), end frequency (f_e), amplitude threshold (th) of normalized spectrum (number from 0 to 1). The spectra of signal samples were calculated and normalized. We took characteristic peaks in this spectrum, considering their width and amplitude. The width was taken with a margin based on the width of the sample peak at the base. The threshold was also taken with a margin based on the peak amplitude. The stock was taken about 25 percent. If recorded signal does not have all the spectral components of the searched object we assume that this object does not exist in the signal.

The second stage of searching algorithm is estimation of cross correlation functions of the recorded signal and the samples of the searched objects which have passed the first stage of the algorithm. Both recorded and sample signals are normalized by amplitude before estimation. We calculate maximum absolute values of the estimated functions (C_m^i) and divide those values to the energies of the corresponding sample signals (E_i):

$$C_e^i = C_m^i / E_i \quad (1)$$

We do it because signals of greater energy have greater correlation despite their form. When we have all C_e^i values we calculate the maximum value $C_e^{\max}$. Then we decide that object (i) exists in the signal when C_e^i satisfies the condition

$$C_e^{\min} < C_e^i < k * C_e^{\max}, \quad (2)$$

where $C_{\min}$ is minimal value and k is tolerance coefficient.

III. RESULTS

In order to check real-time performance of the algorithm we performed tests for three different drones: DJI Phantom 3, DJI Mavic Air, DJI Mavic Mini. Test signals for the database were recorded in the studio in the form of fragments from 120 to 180 seconds long with a sampling rate of 44100 Hz. To simplify testing and analysis we used built-in laptop microphone. Parameters of the recorded sound and the object samples (after resampling) were 8000 Hz, 16 bit, mono. Considering the speed of modern drones up to 65 km / h and the relatively small detection range, it is necessary to choose short time intervals for cyclic detection and recognition. In our case, the time was set to 2.5 seconds. Test recordings were made under city noise conditions for all three drones at various distances while static and while maneuvering. The main results are presented in tab. 1 and 2.

TABLE I. ALGORITHM PARAMETERS

Type of drone	Energy peaks	f_b , Hz	f_e , Hz	Sample energy	th
DJI Phantom 3	first	330	390	0.00386	0.6
	second	510	570		0.3
	third	670	730		0.6
DJI Mavic Air	first	335	365	0.00821	0.6
	second	680	720		0.3

Type of drone	Energy peaks	f_b , Hz	f_e , Hz	Sample energy	th
DJI Mavic Mini	first	335	365	0.00456	0.6
	second	400	430		0.2
	third	680	720		0.2

TABLE II. ALGORITHM PERFORMANCE

Type of drone	Accuracy, %	False negative results, %
DJI Phantom 3	75.75	3
DJI Mavic Air	87.5	9.4
DJI Mavic Mini	61.1	23.3

IV. CONCLUSIONS

The work is devoted to the development and testing of an algorithm for detecting and recognition UAVs under conditions of external noise. The main parameters of the algorithm are presented. It is shown that the recognition accuracy is up to 87.5% at a recording time of 2.5 seconds. The signal with the highest energy appears to have a greater chance of high correlation values. Therefore, the further development of the algorithm is seen in the analysis of not just the maximum CCF value, but its implementation depending on the time shift between the analyzed signal and the sample (frequency, peak width). At the moment, research has been carried out for static positions, the effects associated with the movement of an object (Doppler effect and its own displacement of sound characteristics) are a matter of further research

REFERENCES

- [1] S. Al-Emadi, A. Al-Ali, A. Mohammad and A. Al-Ali, "Audio Based Drone Detection and Identification using Deep Learning," *2019 15th International Wireless Communications & Mobile Computing Conference (IWCMC)*, 2019, pp. 459-464.
- [2] Z. Kaleem and M. H. Rehmani, "Amateur drone monitoring: State-of-the-art architectures, key enabling technologies, and future research directions" *IEEE Wireless Commun.*, vol. 25, no. 2, pp. 150-159, Apr. 2018.
- [3] M. M. Azari, H. Sallouha, A. Chiumento, S. Rajendran, E. Vinogradov and S. Pollin, "Key technologies and system trade-offs for detection and localization of amateur drones" *IEEE Commun. Mag.*, vol. 56, no. 1, pp. 51-57, Jan. 2018.
- [4] I. Guvenc, F. Koohifar, S. Singh, M. L. Sichitiu, and D. Matolak, "Detection tracking, and interdiction for amateur drones," *IEEE Commun. Mag.*, vol. 56, no. 4, pp. 75-81, Apr. 2018.
- [5] J. Lu, J. Yang, X. Liu, G. Liu, and Y. Zhang, "Robust direction of arrival estimation approach for unmanned aerial vehicles at low signal-to-noise ratios," *IET Signal Process.*, vol. 13, no. 4, pp. 456-463, 2019.
- [6] M. Nijim and N. Mantrawadi, "Drone classification and identification system by phenome analysis using data mining techniques," in *Proc. IEEE Symp. Technol. Homeland Secur. (HST)*, May 2016, pp. 1_5. [Online]. Available: <http://ieeexplore.ieee.org/document/7568949/>
- [7] S. Jeon, J.-W. Shin, Y.-J. Lee, W.-H. Kim, Y. Kwon, and H.-Y. Yang, "Empirical study of drone sound detection in real-life environment with deep neural networks," 2017, arXiv:1701.05779. [Online]. Available: <https://arxiv.org/abs/1701.05779>
- [8] A. Bernardini, F. Mangiatordi, E. Pallotti, and L. Capodiferro, "Drone detection by acoustic signature identification," *Electron. Imag.*, vol. 2017, no. 10, pp. 60-64, 2017.